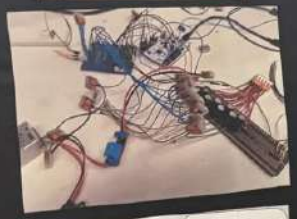
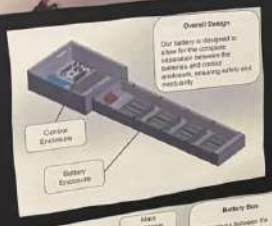
# EE-490 Senior Design Project

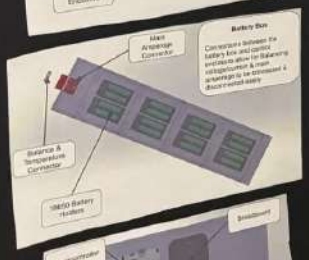
Michael Collins, Dillon Peterson, Ryan Rabideau, Matthew Herber

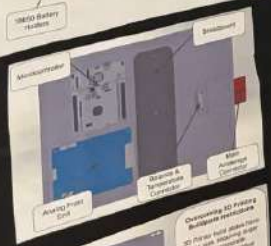
## Design and Assembly Process

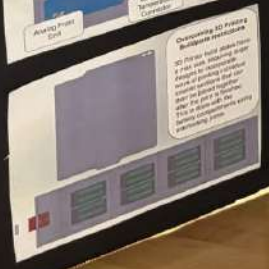




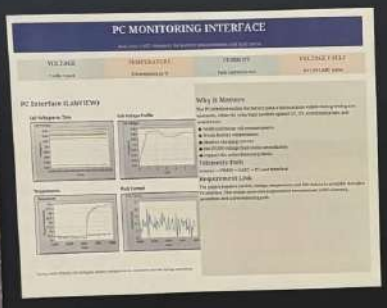


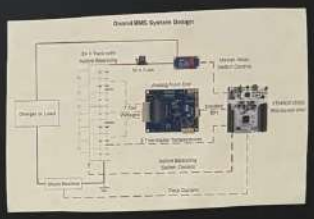



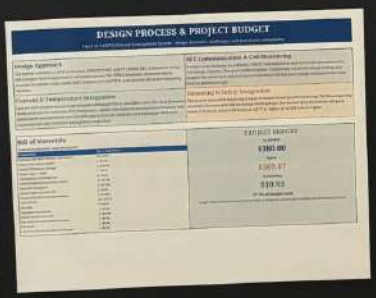




## PC MONITORING INTERFACE

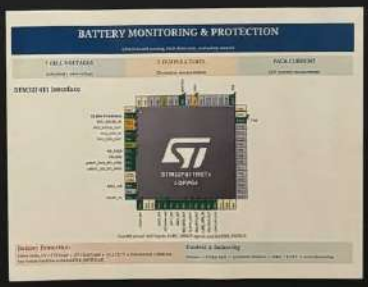



## DESIGN PROCESS & PROJECT BUDGET




## BATTERY MONITORING & PROTECTION



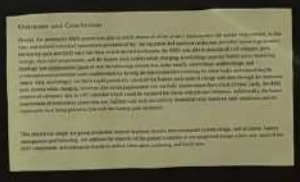





## Microcomputer Software Flowchart

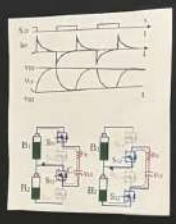


## Conclusion and Conclusion



## Balancing Circuit Topology

The topology we selected is called switched capacitor based active balancing. Charge from a higher SOC cell is used to charge a capacitor, which then is discharged into a lower SOC BATTERY, as can be seen in the diagrams below.



Blue:
Switch Input Voltage

Yellow:
Switching Capacitor Voltage

